

Final Report

Predicting Commercial Success of Steam Games Summary report for Pixel Publishing

Presentation Link: <https://youtu.be/pMJ9FDwwzgs>

I don't receive my client feedback so this report is all based on my own analysis.

Summary

The dataset has 125,855 Steam games but no column that directly says a game did well so I built the label from Estimated owners. I set the line at 50,000+ owners which sits on a real tier boundary and means that the games are actually sold. That makes 11.3% of games count as a success. I removed 1,178 non-game listings (utilities, soundtracks, demos) using genre and name checks. One thing worth mentioning is that release year predicts success really well, but only because owners is a current snapshot. Older games have had more years to reach 50,000 so release year is mostly measuring how long a game has been out not really how good it is. So I dropped it so the model cannot rank a game just by its age.

Client's questions

Q1: Does genre matter? Yes, but on its own it is weak. A logistic regression using only the 12 genre flags gets an AUC of 0.654. When I fit all features together and read each genre adjusted for the rest, genre still holds up. MMO games have about 1.84x the odds of success and Casual sits lowest at 0.50x. So genre carries its own signal. I just don't use it to say whether a game is a success or not.

Q2: Does combining more pre-launch features help? Yes. The same logistic regression on all 19 pre-launch features reaches an AUC of 0.744, about +0.09 over genre alone and its ROC curve sits above the genre-only one across the whole range. I also used Random Forest and it landed in the same place (0.738) but missed most of the real successes, so I kept the logistic as my main model. Precision on the success class is still low (about ~0.22) so I read this as a ranking tool, not a precise yes or no predictor.

What matters most.

The most important to define success are the number of supported languages (by far the largest), being available on Mac, having an age rating, and sitting in a higher-success genre like Action, RPG, or Strategy. The strongest pulls the other way are Casual, being self-published, and Early Access. However these are associations not causes. Also, languages sitting on top probably reflects a bigger budget and localization effort not the languages themselves. I also clustered the game descriptions (TF-IDF and KMeans) into five themes. Action/combat and

multiplayer descriptions succeed most (about 16%) and casual puzzle least (about 6%) which lines up with the genre results rather than adding a new factor.

Recommendation to the Director.

1. Don't pick projects on genre alone. Genre matters, but a genre-only screen is weak (AUC 0.65). Screening on the whole set of pre-launch traits ranks concepts noticeably better (AUC 0.74).
2. Watch the traits that go with success: broad language support, multi-platform availability (like Mac), and a higher-success genre. Self-published, Casual, and Early Access concepts tend to sit lower.
3. Use the model to rank and shortlist concepts, not to auto-approve or reject. Its precision is low, so it flags plenty of games that will not end up succeeding.
4. Must to keep the limits in mind. The success line is 50,000+ owners from a current snapshot so older games still carry some edge and every result here is association, not causation. A trait going with success does not mean forcing it will create success.